

Lecture 2 Exercise Solutions

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1 Important Equations

Probability Mass Function (pmf)

For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$, a probability mass function is a function such that:

1. $f_X(x_i) \geq 0$
2. $\sum_{i=1}^n f_X(x_i) = 1$
3. $f_X(x_i) = P(X = x_i)$

Cumulative Distribution Function (cdf)

The cumulative distribution function of a discrete variable X , denoted as $F(x)$, is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i).$$

It satisfies:

1. $F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f_X(x_i)$
2. $0 \leq F_X(x) \leq 1$
3. if $x \leq y$, then $F_X(x) \leq F_X(y)$

Useful interval properties:

1. $P(x_a < X \leq x_b) = F_X(x_b) - F_X(x_a)$
2. $P(x_a \leq X \leq x_b) = F_X(x_b) - F_X(x_a) + f_X(x_a)$
3. $P(x_a < X < x_b) = F_X(x_b) - F_X(x_a) - f_X(x_b)$
4. $P(x_a \leq X < x_b) = F_X(x_b) - F_X(x_a) + f_X(x_a) - f_X(x_b)$

Mean & Variance

The mean is

$$\mu_x = \sum_i x_i f_X(x_i),$$

And the variance is

$$\sigma_x^2 = \sum_i x_i^2 f_X(x_i) - \mu_x^2.$$

Binomial Distribution

The binomial distribution pmf is given as:

$$f_X(x) = P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad (1)$$

This describes the probability of getting x successes in n independent Bernoulli trials with $p \in [0; 1]$. The mean and variance is given as:

$$\mu_x = n \cdot p, \quad \sigma_x^2 = n \cdot p(1-p)$$

Geometric Distribution

The geometric distribution gives the probability distribution of X number of Bernoulli trials needed to get one success:

$$f_X(x) = P(X = x) = (1-p)^{x-1} p \quad (2)$$

Poisson Distribution

The Poisson distribution tells the probability of X events in a time interval Δt :

$$f_X(x) = P(X = x) = \frac{(\lambda t)^x}{x!} e^{-\lambda t} \quad (3)$$

Where λ is the event intensity with units $[s^{-1}]$.

2 Exercise 3.1.10

Verify that the following function is a probability mass function, and determine the requested probabilities.

$$f(x) = \frac{8}{7} \left(\frac{1}{2} \right)^x, \quad x = 1, 2, 3$$

First to check if the function is a probability mass function the three criteria described in section 1 should be validated. For the first:

$$f(1) = \frac{4}{7}, \quad f(2) = \frac{2}{7}, \quad f(3) = \frac{1}{7}$$

Which upholds since all values are greater than zero. The second criteria is checked:

$$\sum_{i=1}^3 f(x_i) = f(1) + f(2) + f(3) = \frac{4}{7} + \frac{2}{7} + \frac{1}{7} = 1$$

Which is also true. For the third criteria because the total probability of the variable X having a value inside the domain $x_i = 1, 2, 3$ is equal to one then $f(x_i) = P(X = x_i)$.

a. $P(X \leq 1)$

To calculate this probability, the cumulative mass function (cmf) is used. because $x = 1, 2, 3$ there will only be one term in the sum for $x = 1$:

$$P(X \leq 1) = \sum_{x=1} \frac{8}{7} \left(\frac{1}{2}\right)^x = \frac{8}{7} \left(\frac{1}{2}\right)^1 = \frac{8}{14}$$

b. $P(X > 1)$

The cmf function is used again to calculate the probability now with $x = 2, 3$:

$$P(X > 1) = \sum_{x=2,3} \frac{8}{7} \left(\frac{1}{2}\right)^x = \frac{8}{7} \left(\frac{1}{2}\right)^2 + \frac{8}{7} \left(\frac{1}{2}\right)^3 = \frac{3}{7}$$

c. $P(2 < X < 6)$

Since the function is defined for $x = 1, 2, 3$ the only relevant term for this constraint is $x = 3$:

$$P(2 < X < 6) = \sum_{x=3} \frac{8}{7} \left(\frac{1}{2}\right)^x = \frac{8}{7} \left(\frac{1}{2}\right)^3 = \frac{1}{7}$$

d. $P(X \leq 1 \text{ or } X > 1)$

This constraint covers the whole domain where the function is defined. And from definition the probability is then $P(X \leq 1 \text{ or } X > 1) = 1$ because the variable must take a value inside the domain. This can also be proved by calculating the sum:

$$P(X \leq 1 \text{ or } X > 1) = \sum_{x=1,2,3} \frac{8}{7} \left(\frac{1}{2}\right)^x = \frac{8}{7} \left(\frac{1}{2}\right)^1 + \frac{8}{7} \left(\frac{1}{2}\right)^2 + \frac{8}{7} \left(\frac{1}{2}\right)^3 = 1$$

3 Exercise 3.1.11

Verify that the following function is a probability mass function, and determine the requested probabilities.

$$f(x) = \frac{2x+1}{25}, \quad x = 0, 1, 2, 3, 4$$

First to check if the function is a probability mass function the three criteria described in section 1 should be validated. For the first:

$$f(0) = \frac{1}{25}, \quad f(1) = \frac{3}{25}, \quad f(2) = \frac{5}{25}, \quad f(3) = \frac{7}{25}, \quad f(4) = \frac{9}{25}$$

Which upholds since all values are greater than zero. The second criteria is checked:

$$\sum_{i=0}^4 f(x_i) = \frac{1}{25} + \frac{3}{25} + \frac{5}{25} + \frac{7}{25} + \frac{9}{25} = 1$$

Which is also true. For the third criteria because the total probability of the variable X having a value inside the domain $x_i = 0, 1, 2, 3, 4$ is equal to one then $f(x_i) = P(X = x_i)$.

a. $P(X = 4)$

The probability is calculated as:

$$P(X = 4) = \frac{2 \cdot 4 + 1}{25} = \frac{9}{25}$$

b. $P(X \leq 1)$

Here two values fits the constraint which is $x = 0, 1$ meaning the probability is:

$$P(X \leq 1) = \frac{2 \cdot 0 + 1}{25} + \frac{2 \cdot 1 + 1}{25} = \frac{4}{25}$$

c. $P(2 \leq X < 4)$

For this constraint values $x = 2, 3$ fits meaning:

$$P(2 \leq X < 4) = \frac{2 \cdot 2 + 1}{25} + \frac{2 \cdot 3 + 1}{25} = \frac{12}{25}$$

d. $P(X > -10)$

Since the whole domain $x = 0, 1, 2, 3, 4$ is within this constraint then the solution is $P(X > -10) = 1$.

4 Exercise 3.2.8

Verify that the following function is a cumulative distribution function and, determine the probability mass function and the requested probabilities.

$$F(x) = \begin{cases} 0, & x < -10 \\ 0.25, & -10 \leq x < 30 \\ 0.75, & 30 \leq x < 50 \\ 1, & 50 \leq x \end{cases}$$

First to verify if the function is a cdf the properties in section 1 is checked. Condition two is met since the function is defined within $0 \leq F(x) \leq 1$. Condition 3 is also met since:

$$P(X < -1) < P(-10 \leq X < 30) < P(30 \leq X < 50) < P(50 \leq X).$$

Now to find the pmf which will used in later calculations we first recall the definition:

$$f(x) = P(X = x) = F_X(x) - F_X(x^-)$$

Where $F_X(x^-)$ means we look at the limit of the cdf at x coming from the left. Meanning the pmf can be seen as a jump in the cdf. Looking at the boundary values:

$$F(-10^-) = 0, F(-10) = 0.25 \rightarrow f(-10) = 0.25$$

$$F(30^-) = 0.25, F(30) = 0.75 \rightarrow f(30) = 0.5$$

$$F(50^-) = 0.75, F(50) = 1 \rightarrow f(50) = 0.25$$

And for all other x the pmf function will have a value of zero since there is no jump in the cdf:

$$f(x) = \begin{cases} 0.25, & x = -10 \\ 0.5, & x = 30 \\ 0.25, & x = 50 \\ 0, & \text{otherwise} \end{cases}$$

a. $P(X \leq 50)$

From the definition of F , we have

$$P(X \leq 50) = F(50) = 1.$$

b. $P(X \leq 40)$

Since 40 lies in the interval $30 \leq x < 50$,

$$P(X \leq 40) = F(40) = 0.75.$$

c. $P(40 \leq X \leq 60)$

We use

$$P(40 \leq X \leq 60) = F(60) - F(40) - f(40).$$

As 40 lies in a flat interval $f(40) = 0$. Thus

$$P(40 \leq X \leq 60) = 1 - 0.75 - 0 = 0.25.$$

d. $P(X < 0)$

Since $-10 < x < 30$ corresponds to $F(x) = 0.25$,

$$P(X < 0) = F(0) = 0.25.$$

e. $P(0 \leq X < 10)$

Because $0 \leq x < 30$ is a flat interval,

$$f(10) = 0, \quad P(0 \leq X < 10) = F(10) - F(0) + f(10) = 0.$$

f. $P(-10 < X < 10)$

We compute

$$P(-10 < X < 10) = F(10) - F(-10) - f(10).$$

Since $F(10) = F(-10) = 0.25$ and $f(10) = 0$,

$$P(-10 < X < 10) = 0.$$

5 Exercise 3.3.1

If the range of X is the set $0, 1, 2, 3, 4$ and $f(X = x) = 0.2$, determine the mean and variance of the random variable.

The mean value is found as:

$$\mu_x = \sum_i x_i f_X(x_i) = \sum_i x_i \cdot 0.2 = 0 \cdot 0.2 + 1 \cdot 0.2 + 2 \cdot 0.2 + 3 \cdot 0.2 + 4 \cdot 0.2 = 2$$

And the variance as:

$$\sigma_x^2 = \sum_i x_i^2 f_X(x_i) - \mu_x^2 = 0^2 \cdot 0.2 + 1^2 \cdot 0.2 + 2^2 \cdot 0.2 + 3^2 \cdot 0.2 + 4^2 \cdot 0.2 - 4 = 2 = 2$$

6 Exercise 3.3.7

The range of the random variable X is $[0, 1, 2, 3, x]$ where x is unknown. If each value is equally likely and the mean of X is 6, determine x .

Since each value is equally likely then the pmf function with 5 possible values is:

$$f_X(x) = \frac{1}{5} = 0.2$$

Because the mean is known we can isolate for x :

$$\mu = 0 \cdot 0.2 + 1 \cdot 0.2 + 2 \cdot 0.2 + 3 \cdot 0.2 + x \cdot 0.2 \Rightarrow x = \frac{\mu - 0.2 - 0.4 - 0.6}{0.2}$$

Inserting gives:

$$x = \frac{6 - 1.2}{0.2} = 24$$

7 Exercise 3.4.1

Assume that the wavelength of photosynthetically active radiations (PAR) are uniformly distributed at integer nanometers in the red spectrum from 675 to 700 nm.

a. What are the mean and variance of the wavelength distribution for

this radiation?

The mean and variance are found for a uniform distribution with lower limit a and upper limit b using eq. 3.6 from the book:

$$\mu_\lambda = \frac{b + a}{2} = \frac{675 \text{ nm} + 700 \text{ nm}}{2} = 687.5 \text{ nm}$$
$$\sigma_\lambda^2 = \frac{(b - a + 1)^2 - 1}{12} = \frac{(700 \text{ nm} - 675 \text{ nm} + 1)^2}{12} = 56.25 \text{ nm}^2$$

b. If the wavelengths are uniformly distributed at integer nanometers from 75 to 100 nanometers, how do the mean and variance of the wavelength distribution compare to the previous part? Explain

The mean and variance is calculated using the same equations as before:

$$\mu_\lambda = \frac{75 \text{ nm} + 100 \text{ nm}}{2} = 87.25 \text{ nm}$$
$$\sigma_\lambda^2 = \frac{(100 \text{ nm} - 75 \text{ nm} + 1)^2 - 1}{12} = 56.25 \text{ nm}^2$$

Because the values has shifted compared to a. the mean has also. However because the range is still 25 nm with the same uniform distribution the variance is the same.

8 Exercise 3.5.2

Let X be a binomial random variable with $p = 0.1$ and $n = 10$. Calculate the following probabilities.

a. $P(X \leq 2)$

We use the pmf for the binomial distribution (eq. 1) and calculate the probability as:

$$P(X \leq 2) = f_X(0) + f_X(1) + f_X(2) = 0.93$$

This can be computed using "binopdf(x,n,p)" in MATLAB.

b. $P(X > 8)$

Using the same equation as before however now $x = 9, 10$:

$$P(X > 8) = f_X(9) + f_X(10) = 9 \cdot 10^{-9}$$

c. $P(X = 4)$

$$P(X = 4) = f_X(4) = 0.0112$$

d. $P(5 \leq X \leq 7)$

$$P(5 \leq X \leq 7) = f_X(5) + f_X(6) + f_X(7) = 0.00163$$

9 Exercise 3.5.4

The random variable X has a binomial distribution with $n = 10$ and $p = 0.5$.

Sketch the probability function of X .

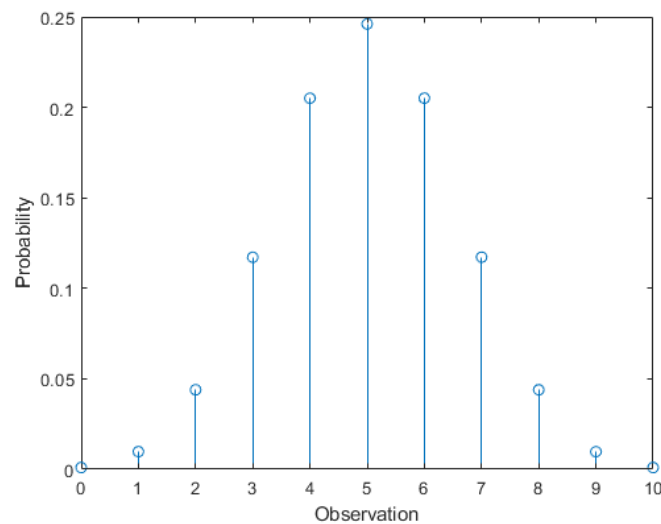


Figure 1: Probability function of X with $n = 10$ and $p = 0.5$.

a. What value of X is most likely?

From the plot in fig. 1 the most likely value is $x = 5$ where $P(X = 5) = 0.246$.

b. What value(s) of X is least likely?

From the plot in fig. 1 the least likely values are $x = 0$ and $x = 10$ where $P(X = 0) = P(X = 10) = 0.000977$.

c. Repeat the previous parts with $p = 0.01$.

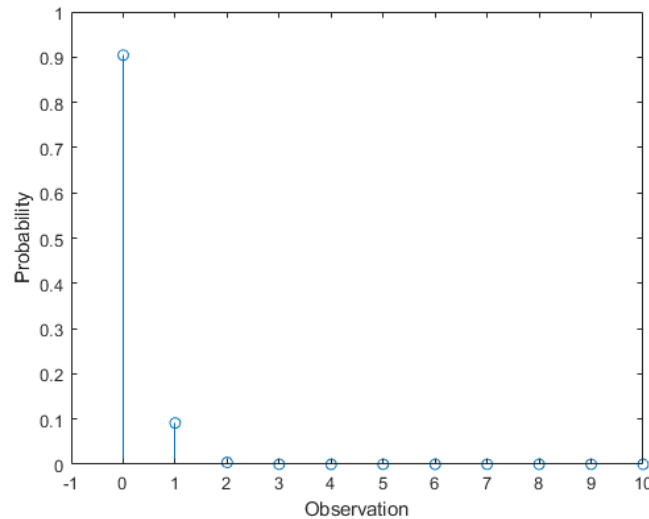


Figure 2: Probability function of X with $n = 10$ and $p = 0.01$.

From fig. 2 the most and least likely values are:

$$P(X = 0) = 0.904, \quad P(X = [3; 10]) = 0$$

10 Exercise 3.5.13

Because all airline passengers do not show up for their reserved seat, an airline sells 125 tickets for a flight that holds only 120 passengers. The probability that a passenger does not show up is 0.1, and the passengers behave independently.

a. What is the probability that every passenger who shows up can take the flight?

In order for all passengers showing up to be able to take the flight 5 passengers or

more must not show up. This can also be written as one minus 4 or less passengers not showing up:

$$P(X \geq 5) = 1 - P(X \leq 4) = 1 - [f_X(0) + f_X(1) + f_X(2) + f_X(3) + f_X(4)]$$

This can be solved using "1-binocdf(4,125,0.1)" in MATLAB giving:

$$P(X \geq 5) = 1 - P(X \leq 4) = 0.9961$$

Meaning there is a 99.61% probability that every passenger that shows up can take the flight.

b. What is the probability that the flight departs with empty seats?

In order for the flight to depart with empty seats more than 5 passengers need to not show up:

$$P(X > 5) = 1 - P(X \leq 5) = 1 - [f_X(0) + f_X(1) + f_X(2) + f_X(3) + f_X(4) + f_X(5)]$$

Solving again using MATLAB gives:

$$P(X > 5) = 0.9886$$

11 Exercise 3.6.1

Suppose that the random variable X has a geometric distribution with $p = 0.5$. Determine the following probabilities.

a. $P(X = 1)$

This can be solved using the geometric distribution pmf defined in eq. 2 and using the given value of $p = 0.5$:

$$P(X = 1) = f_X(1) = (1 - 0.5)^{1-1} \cdot 0.5 = 0.5$$

b. $P(X = 4)$

$$P(X = 4) = f_X(4) = (1 - 0.5)^{4-1} \cdot 0.5 = 0.0625$$

c. $P(X = 8)$

$$P(X = 8) = f_X(8) = (1 - 0.5)^{8-1} \cdot 0.5 = 0.0039$$

d. $P(X \leq 2)$

Here we should sum over $x = \{1, 2\}$ with the pmf:

$$P(X \leq 2) = f_X(1) + f_X(2) = (1 - 0.5)^{1-1} \cdot 0.5 + (1 - 0.5)^{2-1} \cdot 0.5 = 0.75$$

e. $P(X > 2)$

This is the inverse problem of d. meaning:

$$P(X > 2) = 1 - P(X \leq 2) = 1 - 0.75 = 0.25$$

12 Exercise 3.8.2

The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour. Determine the following probabilities:

a. exactly 5 calls in one hour

This is solved using the Poisson pmf defined in eq. 3 with a phone call intensity of $\lambda = 10$ calls/hour and the time span of $t = 1$ hour:

$$P(X = 5) = f_X(5) = \frac{(\lambda t)^x}{x!} e^{-\lambda t} = \frac{(10 \cdot 1)^5}{5!} e^{-10 \cdot 1} = 0.0378$$

b. 3 or fewer calls in one hour

To solve this the pmf is again used but now summing over $X = \{0, 1, 2, 3\}$:

$$P(X \leq 3) = \sum_{i=0}^3 f_X(x_i) = \sum_{i=0}^3 \frac{(10 \cdot 1)^{x_i}}{x_i!} e^{-10 \cdot 1} = 0.0103$$

c. exactly 15 calls in two hours

We use the same equation as in a. but now $X = 15$ and $t = 2$ hours:

$$P(X = 15) = f_X(15) = \frac{(10 \cdot 2)^{15}}{15!} e^{-10 \cdot 2} = 0.052$$

d. exactly 5 calls in 30 minutes

To find the probability now $X = 5$ and $t = 0.5$ hours:

$$P(X = 5) = f_X(5) = \frac{(10 \cdot 0.5)^5}{5!} e^{-10 \cdot 0.5} = 0.175$$